

## TECHNICAL DATA SHEET (TDS) – PRODUCT 3 (REVISED)

**PRODUCT NAME:** High-Optical Grade Anti-Yellowing Tail Light Lens Pair

**APPLICATION:** BMW E38 7-Series Rear Light Assembly (Facelift & Pre-facelift)

**ORIENTATION:** Left Hand (LH) & Right Hand (RH) Mirror-Symmetric Pairs

**OEM LH CROSS-REFERENCE:** 63218360079, 82199402991,  
63218381245, 63218387637

**OEM RH CROSS-REFERENCE:** 63218360080, 82199402992,  
63218381246, 63218387638

### 1. DIMENSIONS & GEOMETRY

- **Wall Thickness:** 3.00 mm constant wall section ( $\pm 0.05$  mm mechanical injection tolerance).
- **Edge Interface:** Precision co-molded perimeter groove for seamless Integration with the housing and backing seal.
- **Color Segregation:** Aspherical clear crystal sector (Reverse light) / Deep amber or smoked red sector (Indicator/Tail).

### 2. MATERIAL SPECIFICATIONS

- **Polymer Base:** High-Flow Optical-Grade PMMA (Polymethyl Methacrylate / Acrylic) according to ISO 24025.
- **Additive Package:** Fortified with specialized hindered amine light stabilizers (HALS) and anti-UV masterbatch.
- **Surface Coating (Critical Shield):** Polisziloxán-alapú UV-védő keménybevonat (Siloxane-based Hard Coat) via dip-coating process, thermally or UV cured.

### 3. WEATHERING & LIFE EXPECTANCY

- **Accelerated Aging Test:** Must pass a **2000-hour Xenon-Arc Weathering Test** according to ISO 4892-2 (Requirement: **Delta E < 1.0**, absolute zero yellowing, hazing, or chalking of the clear reverse section under extreme UV index).
- **Impact Resilience:** High Charpy notched impact strength according to ISO 179 to resist road debris pitting.

